

Solution: By Definition, y will be in $\text{Span}\{v_1, v_2, v_3\}$ if y is a linear combination of the vectors v_1, v_2 and v_3 , i.e.

$x_1 v_1 + x_2 v_2 + x_3 v_3 = y$ for some $x_1, x_2, x_3 \in \mathbb{R}$. This is only possible if there exists a solution set of the augmented matrix

$$\begin{bmatrix} v_1 & v_2 & v_3 & y \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ -2 & -7 & 0 & h \end{bmatrix} \xrightarrow[\text{Row 2} + \text{Row 1}]{\text{Row 2} =}$$

$$\begin{bmatrix} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ -2 & -7 & 0 & h \end{bmatrix} \xrightarrow[\text{Row 3} + 2 \times \text{Row 1}]{\text{Row 3} =}$$

$$\begin{bmatrix} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & -5h \end{bmatrix} \xrightarrow[\text{Row 3} - 3 \times \text{Row 2}]{\text{Row 3} =}$$

Verification of the final matrix in echelon form

1. There are no zero rows in the matrix. $\therefore$ The condition that the zero rows must be below the non-zero rows is vacuously true.

2. Row 1: Leading element $\rightarrow$ Column no.: 1

Row 2: Leading element $\rightarrow$ Column no.: 2

Row 3: Possible leading element $\rightarrow h-5$ Column no.: 4

$\therefore$ The condition that the leading element of a row must be in a column to the right of the columns of the leading elements of the rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

$\therefore$ The condition that for each pivot column, the rows below the pivot position contain 0 is True.

$\therefore$ The matrix $\begin{bmatrix} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & h-5 \end{bmatrix}$ is in echelon form.

There exists a row of the form $[0 \ 0 \ 0 \ \dots \ 0 \ b]$.